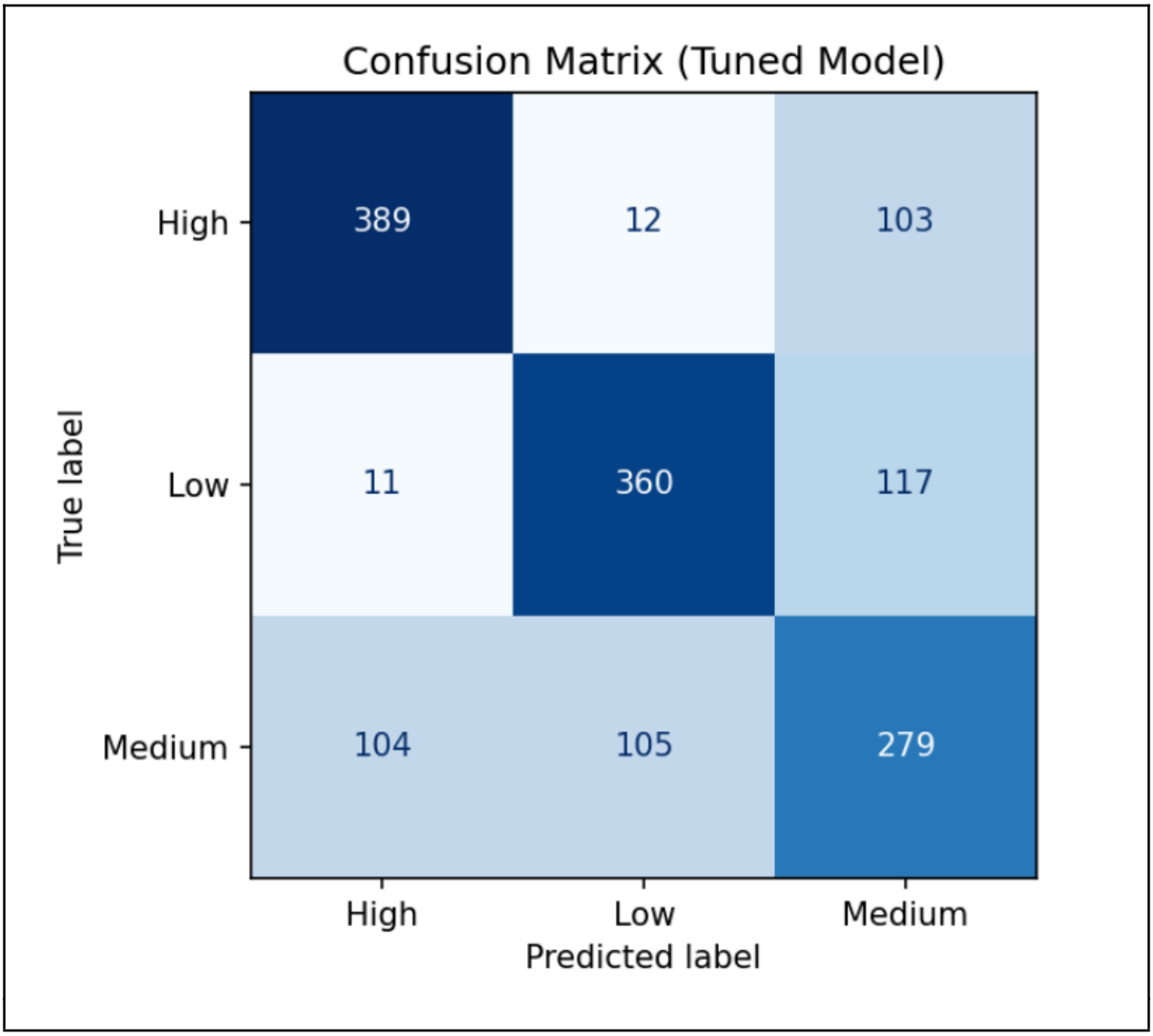
Experiment 5

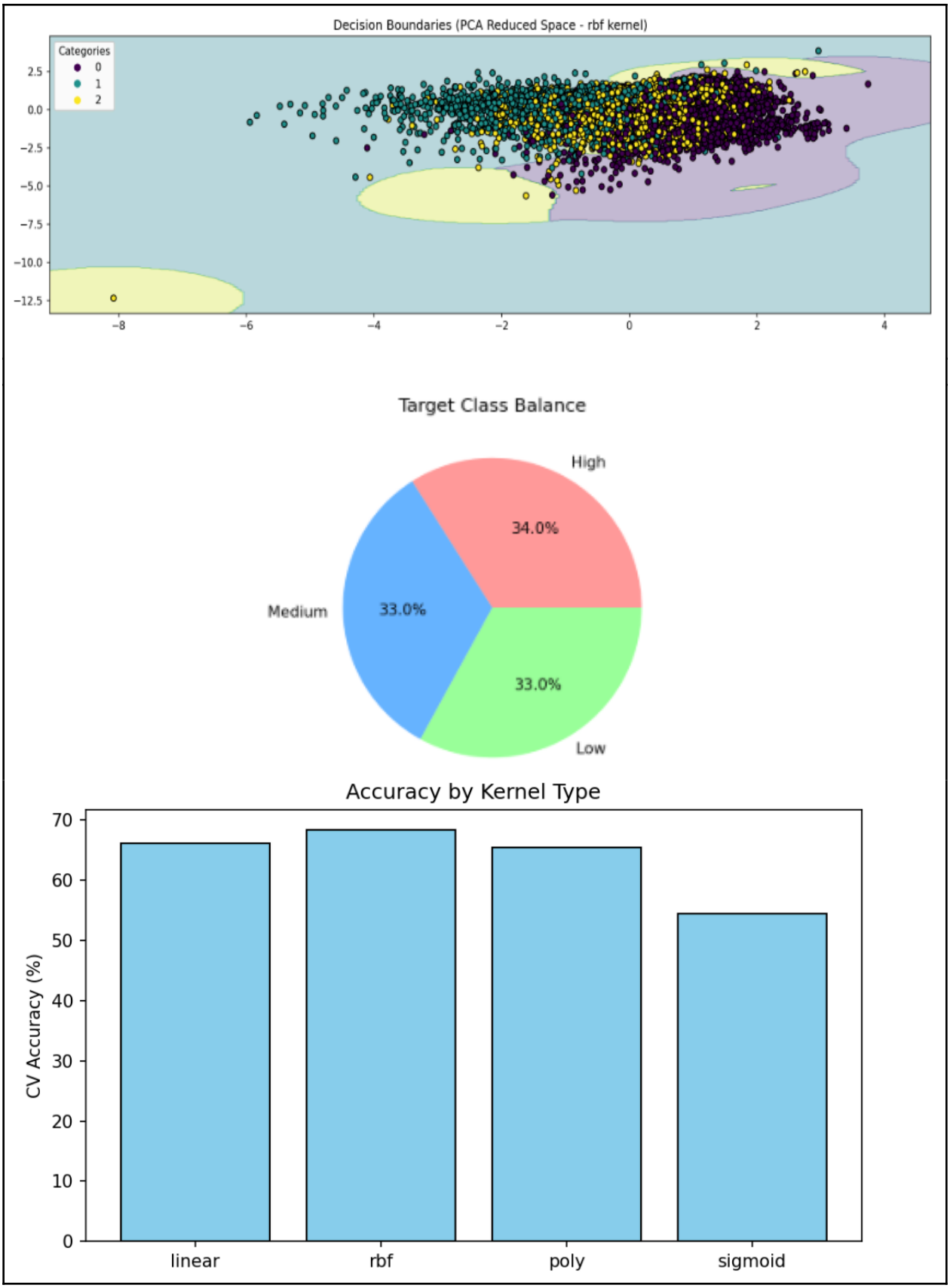
Aim : Implement Support Vector Machine (SVM) for classification with hyperparameter tuning

Data Source : (Used Car Price Prediction Dataset)

- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- [Dataset Link]

Output :




Decision Boundaries (PCA Reduced Space - rbf kernel)
Categories
0
1
2
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−8
−6
−4
−2
0
2
4
Target Class Balance
High
34.0%
Medium
33.0%
33.0%
Low
Accuracy by Kernel Type
CV Accuracy (%)
70
60
50
40
30
20
10
0
linear
rbf
poly
sigmoid

Conclusion :
The implementation of Support Vector Machine (SVM) for classification demonstrates how machine learning can effectively separate different classes using an optimal hyperplane with maximum margin. By applying appropriate kernel functions, SVM can handle both linearly and non-linearly separable data. Hyperparameter tuning plays a crucial role in improving the model's performance by selecting the best values for parameters such as **C, gamma, and kernel type**. Proper tuning results in better classification accuracy and improved generalization on unseen data, making SVM a powerful and widely used algorithm for classification tasks.